

MODEL QUESTION PAPER

TED(15)- 6245
(REVISION-2015)

Reg. No.

Signature

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/MANAGEMENT/COMMERCIAL PRACTICE BIOPHOTONICS

(Maximum Marks: 100)

TIME: 3HRS

PART – A

(Maximum Marks: 10)

Answer *all Questions* in one or two sentences. Each question carries **2 marks**.

I.

1. Define the term monochromaticity of laser?
2. List the different gain medium of laser?
3. Outline any two applications of Nd-YAG laser in medical field?
4. List any four advantages of optical biosensor?
5. Give the names of the laser used in ophthalmology?

(5x2=10 marks)

PART – B

(Maximum Marks: 30)

Answer *any FIVE* of the following questions. Each question carries **6 marks**.

II.

1. Explain the optical properties of living tissue?
2. Explain the process of Q switching and its requirements?
3. Explain CO2 laser with schematic?
4. Describe the basic principle of optical biosensors?
5. Explain the electrostatic and mechanical sorting devices?
6. List any six application of laser in dermatology?
7. Discuss about the limitation of photodynamic therapy?

(5x6=30 marks)

PART – C

(Maximum Marks: 60)

Answer **one full question** from each Unit. Each full question carries **15 marks**.

UNIT I

- III. . a)** Explain the properties of lasers? (8)
b) Discuss about laser induced changes in tissue? (7)

OR

- IV. a)** Explain on ,
(i) Stimulated emission
(ii) Saturable absorber
(iii) Temporal coherence [3x5=15]

UNIT II

- V. a)** Explain the construction of Ruby laser with schematic? (8)
b) Explain the main elements of laser construction? (7)

OR

- VI. a)** Illustrate the construction of He-Ne laser? (8)
b) Explain the operation of Nd-YAG laser? (7)

UNIT III

- VII. a)** Explain the block diagram of flow cytometry ? (8)
b) Discuss the need of bioimaging? (7)

OR

- VIII. a)** Explain In vivo imaging with example? (8)
b) Give an account on cellular imaging? (7)

UNIT IV

- IX. a)** Explain about photosensitive dyes and glues? (7)
b) Discuss about photosensitisation of cells in photodynamic therapy? (8)

OR

- X. a)** Explain how vascular lesions is treated with laser? (8)
b) Discuss about the application of laser in ophthalmology? (7)

(4 x 15 = 60 marks)

